E5 Wind turbine model

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Introduction

- In this exercise you will simulate a 1-4 wind turbines in a conventionally neutral boundary layer.
- The wind farm consists of max. 4 wind turbines.
- The IEA 15 MW Reference wind turbine is used, it has:
  - a rotor diameter of D = 240 m,
  - a hub height of h = 150 m and
  - a rated power of 15 MW
- Cyclic boundary conditions are applied in x- and y-direction, so that the wind farm is effectively infinitely large.

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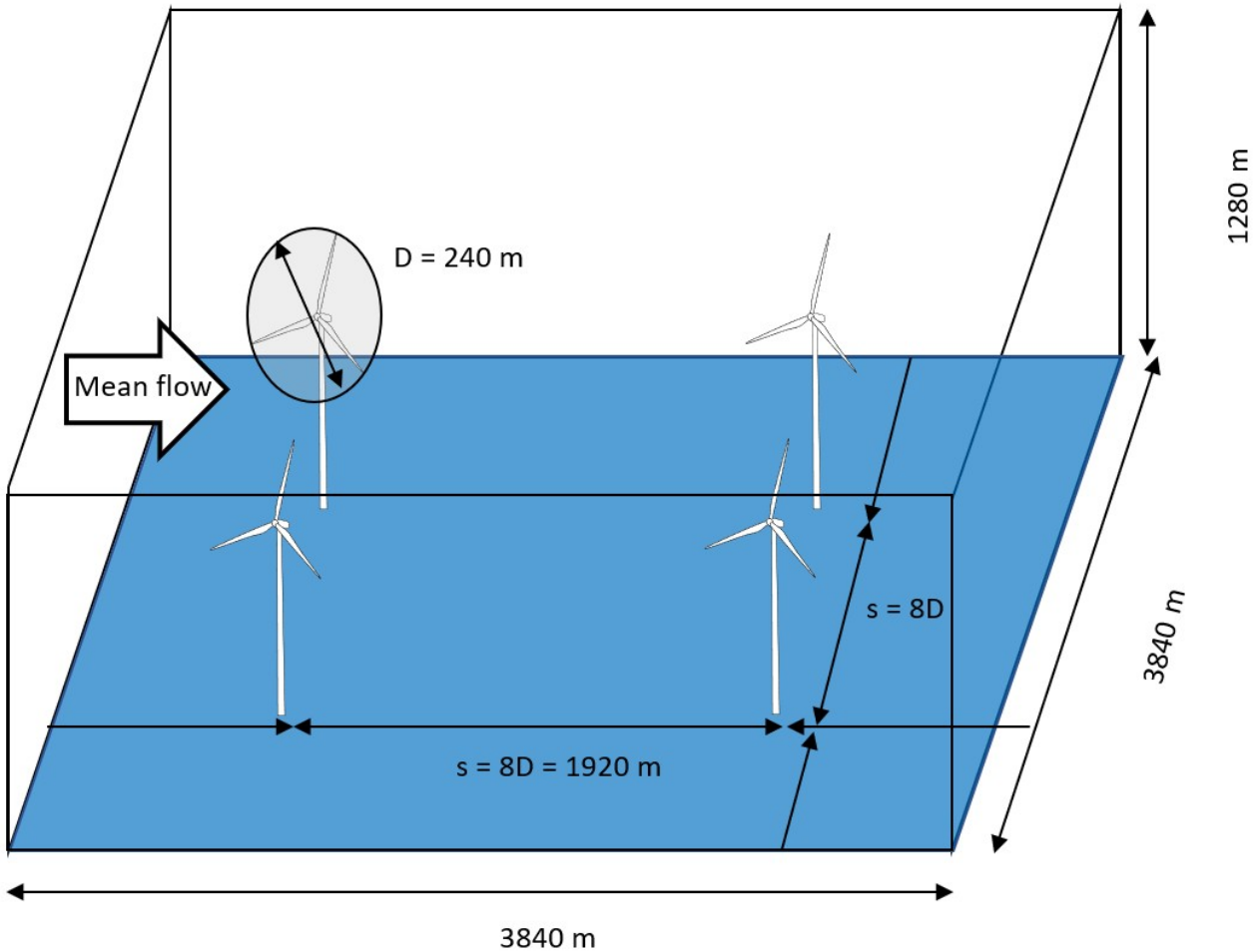
Setup with 4 turbines



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Setup

- **Physics**
  - Flow is initialized with vertically constant velocity profiles, corresponding to the geostrophic wind: u(z) = ug_surface, v(z) = vg_surface
  - initial potential temperature profile is constant up to z = 500 m and has a lapse rate of +0.35 K/(100 m) above
  - Flow is driven by a geostrophic wind of ug, vg = 12.0, -3.8 m/s
    - This results in a wind direction at hub height that is approximately aligned with the x-axis after 2 h.
  - No surface heating or cooling → neutral boundary layer
- **Numerics**
  - grid spacing is 40 m → 6 grid points per rotor diameter
  - cyclic boundary conditions along x and y → infinite wind farm
  - Data output of instantaneous and averaged data every 30 minutes
- **Wind turbines**
  - 4 equally spaced wind turbines, spacing 8 rotor diameters = 1920 m
  - hub height = 150.0 m, Rotor diameter = 240 m
  - active pitch and yaw control

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How to start the simulation

- The wind turbine model requires NetCDF version 4. Do the following steps in your installation directory:
  - create a copy of the `.palm.config.default` file and rename it to: `.palm.config.default_netcdf4`
  - In this file add „ `-D__netcdf4“ to the „%cpp_options …“`
  - recompile palm using the new config: `palmbuild -c default_netcdf4`

- Provide the input files
  - create a new directory "`e5_wtm_exercise/INPUT`" in your "`JOBS`" directory
  - copy the input files `e5_wtm_exercise_p3d` and `e5_wtm_exercise_wtm` to the `INPUT` directory

- Start the simulation using the new configuration file (adapt if you have more cores available!):
  - `palmrun -r e5_wtm_exercise -c default_netcdf4 -a “d3#” -X 4 -T 4`

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Questions

Part 1: wind park simulation

1. Is the mean flow in a steady state at the end of the simulation (t = 2 h)?
   - use: `plot_time_series.py, plot_profiles.py`
2. Describe the mean velocity field at hub height, especially near the turbines. Explain how it is related to the pressure field.
   - use: `plot_xy_av.py`
3. Are the wakes rotating? In which direction and why?
   - use: `plot_xy_av.py, plot_xz_av.py`

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Questions & Tasks

Part 2: single wind turbine & wake interaction

4. Reduce the set-up to a single turbine located at the center of the domain. Vary the geostropic wind speed in and try to reproduce the power curve of the turbine!

   Hints:

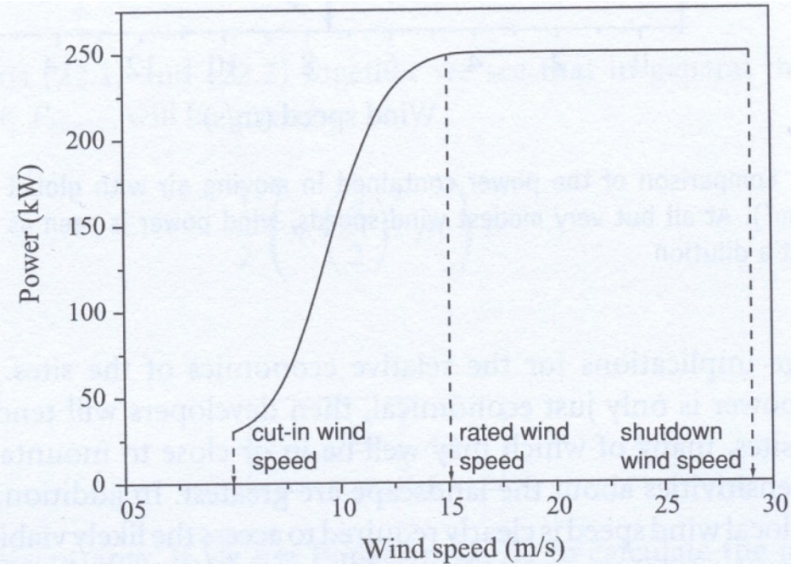
   - you can set `vg_surface = 0.0` for a single turbine set-up
   - you might use MS Excel/LO Spreadsheet for that purpose

5. Place a 2nd turbine in the wake of the first one using different spacings:

   2 D, 5 D, 8 D

   How do the turbines interact regarding wind speed recovery?

   What distance between turbines do you recommend based on these lightweight simulations?



Example of a power curve

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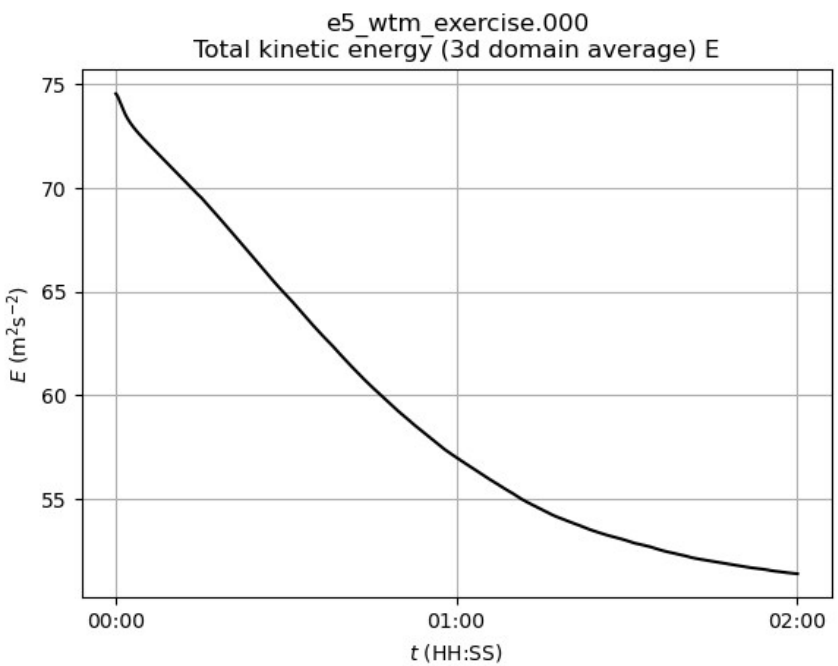
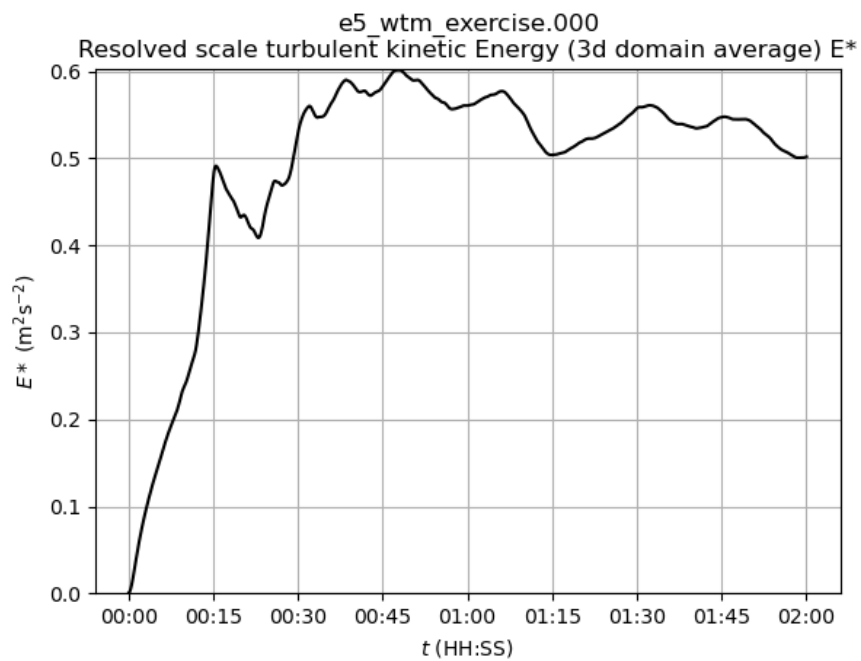
Answers

1. Is the mean flow in a steady state at the end of the simulation (t = 2 h)?

- No. It seems that the total kinetic energy E has reached a steady state, but it is only a minimum of the inertial oscillation.
- The inertial oscillation has a period of ~15 h at a latitude of +-55°





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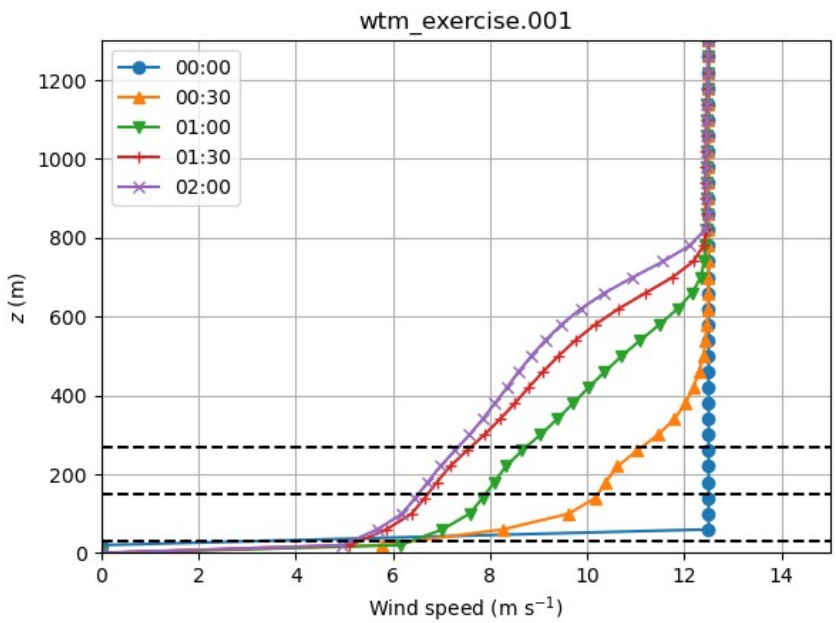
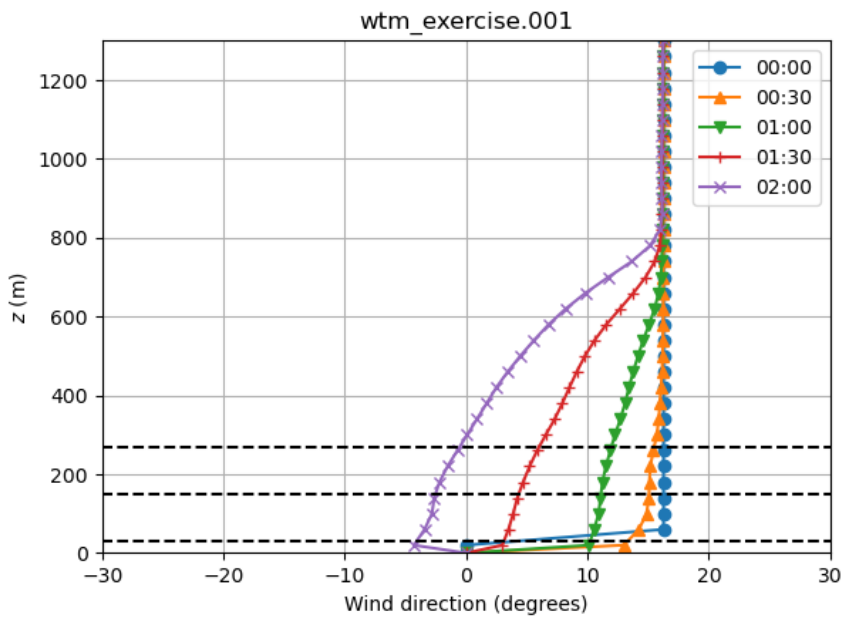
Answers

- The u-component is nearly steady at t = 2 h, but the v-component changes.
- A longer run will take too much time for a seminar exercise.
- If averaging is done over a relatively short period of time (here 30 minutes), the flow can be assumed to be quasi-stationary. But averaged results have to be interpreted with care.





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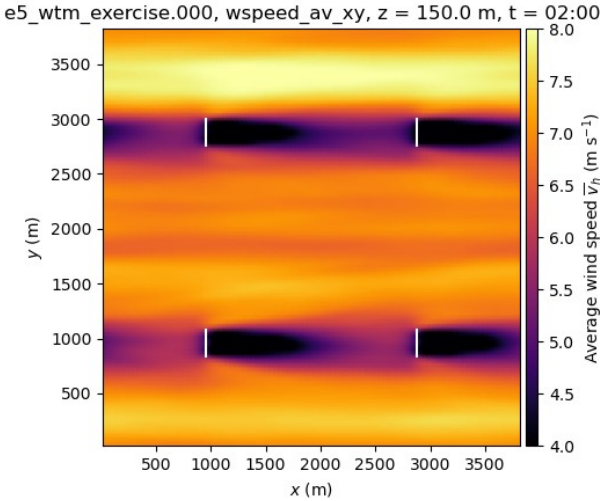
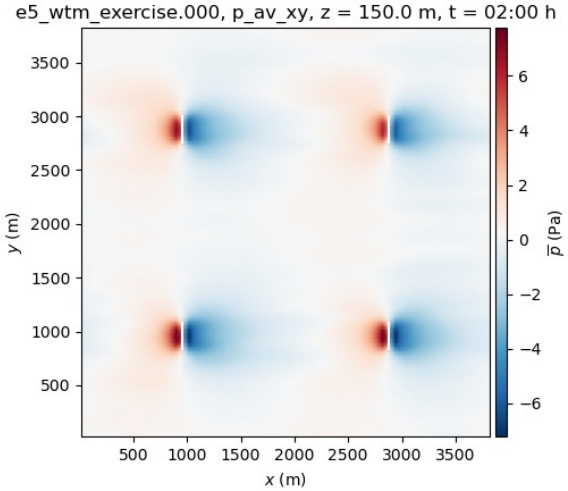
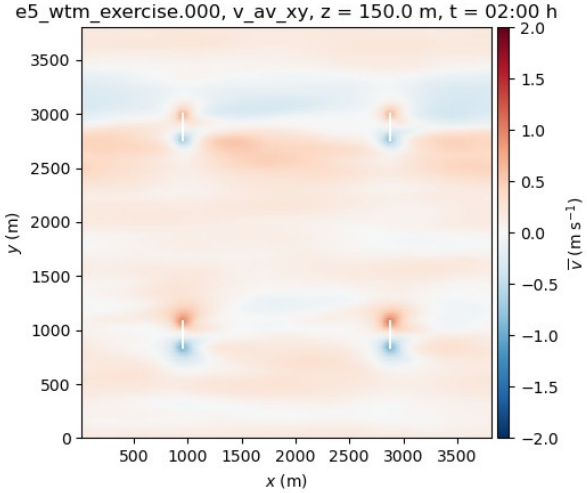
Answers

2. Describe the mean velocity field at hub height, especially near the turbines. Explain how it is related to the pressure field.

- Decreased wind speed behind the wind turbines (= wake)
  - deceleration caused by positive pressure gradient upstream and downstream of the rotor disc
- Higher wind speed between the turbine rows
- Lateral deflection of the flow (+-v), (flow divergence) at the rotor edges.







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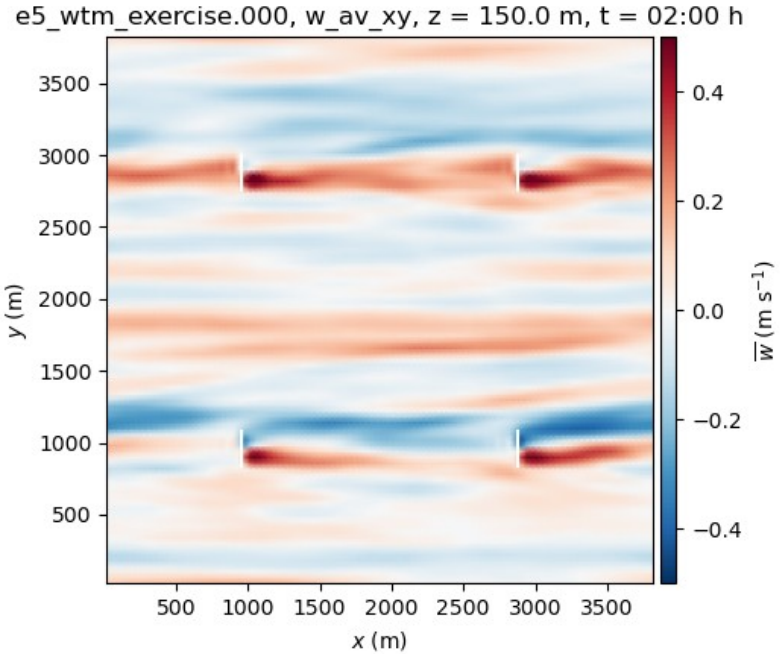
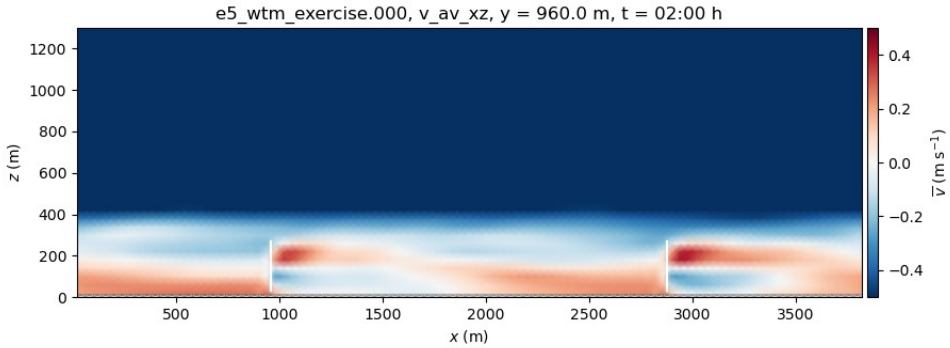
Answers

3. Are the wakes rotating? In which direction and why?

- Yes, they rotate counterclockwise around the positive x-axis.
- Why? The rotor rotates clockwise. The opposite sense of direction is the result of conservation of angular momentum: The flow applies clockwise torque on the rotor and the rotor applies counterclockwise torque on the flow.



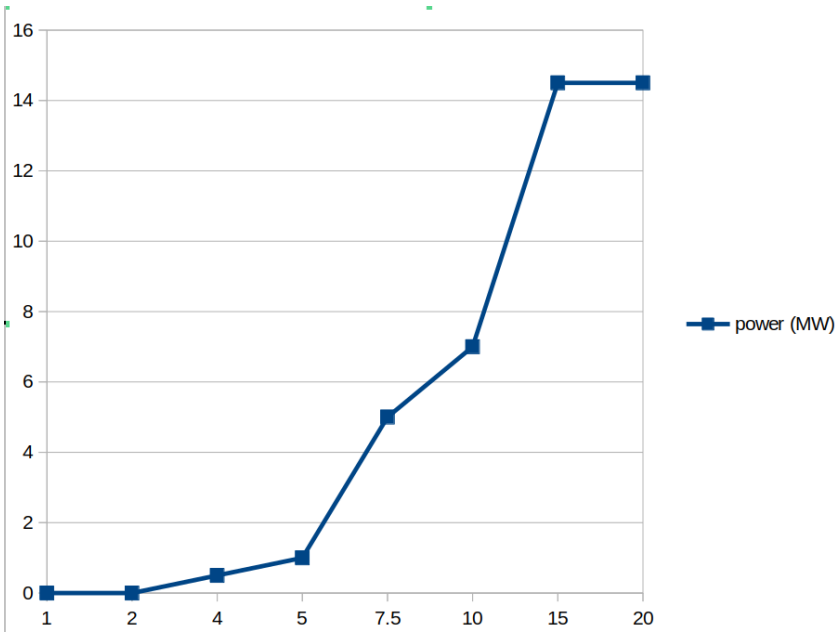


E5: Wind turbine model

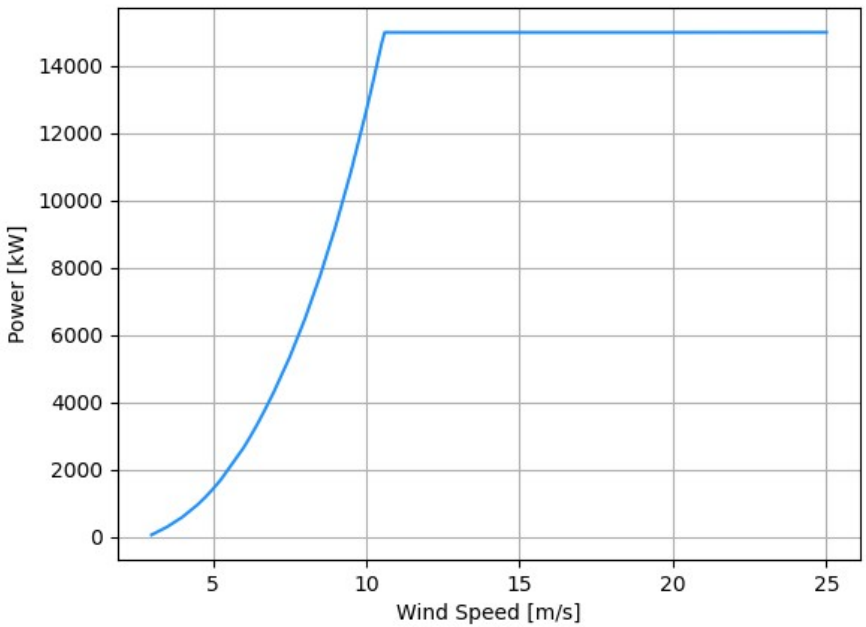
Answers

4. Reproduce the power curve of the turbine

Power curve from PALM



IEA 15 MW power curve



E5: Wind turbine model

Answers

4. Reproduce the power curve of the turbine

IEA_15MW_240_RWT

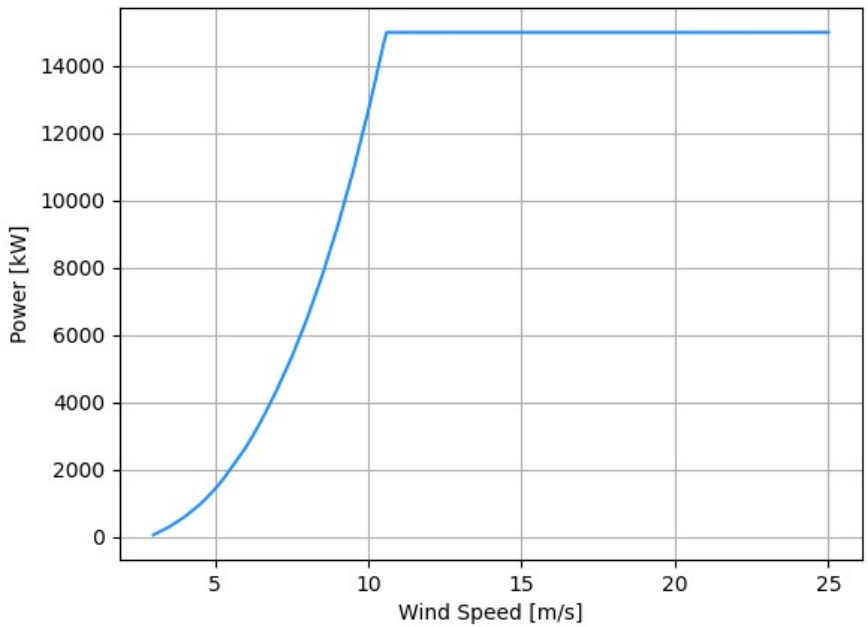
Link to Tabular Data

A .csv file is available on GitHub.

Key Parameters

Item	Value	Units
Name	IEA 15 MW RWT	N/A
Rated Power	15000	kW
Rated Wind Speed	10.6	m/s
Cut-in Wind Speed	3	m/s
Cut-out Wind Speed	25	m/s
Rotor Diameter	240	m
Hub Height	150	m
Drivetrain	Direct Drive	N/A
Control	Pitch Regulation	N/A
IEC Class	1B	N/A

IEA 15 MW power curve

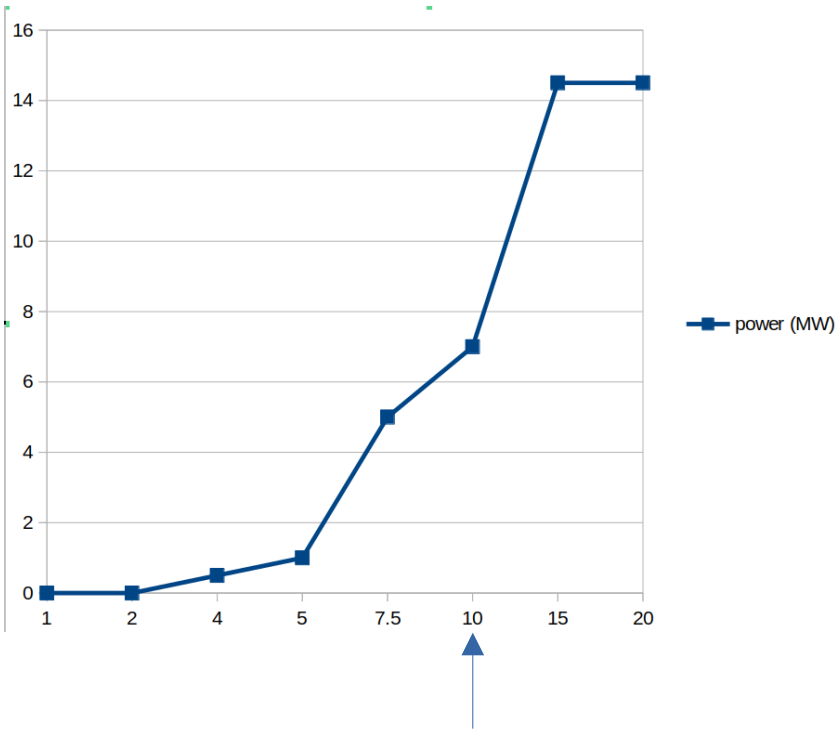


E5: Wind turbine model

Answers

4. Reproduce the power curve of the turbine

Power curve from PALM



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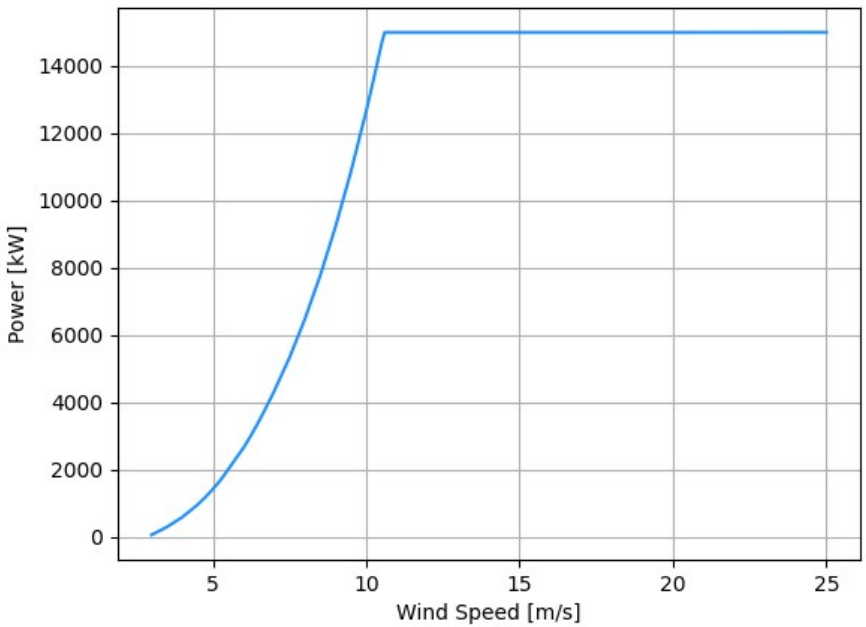

Geostrophic wind, not wind at hub height!

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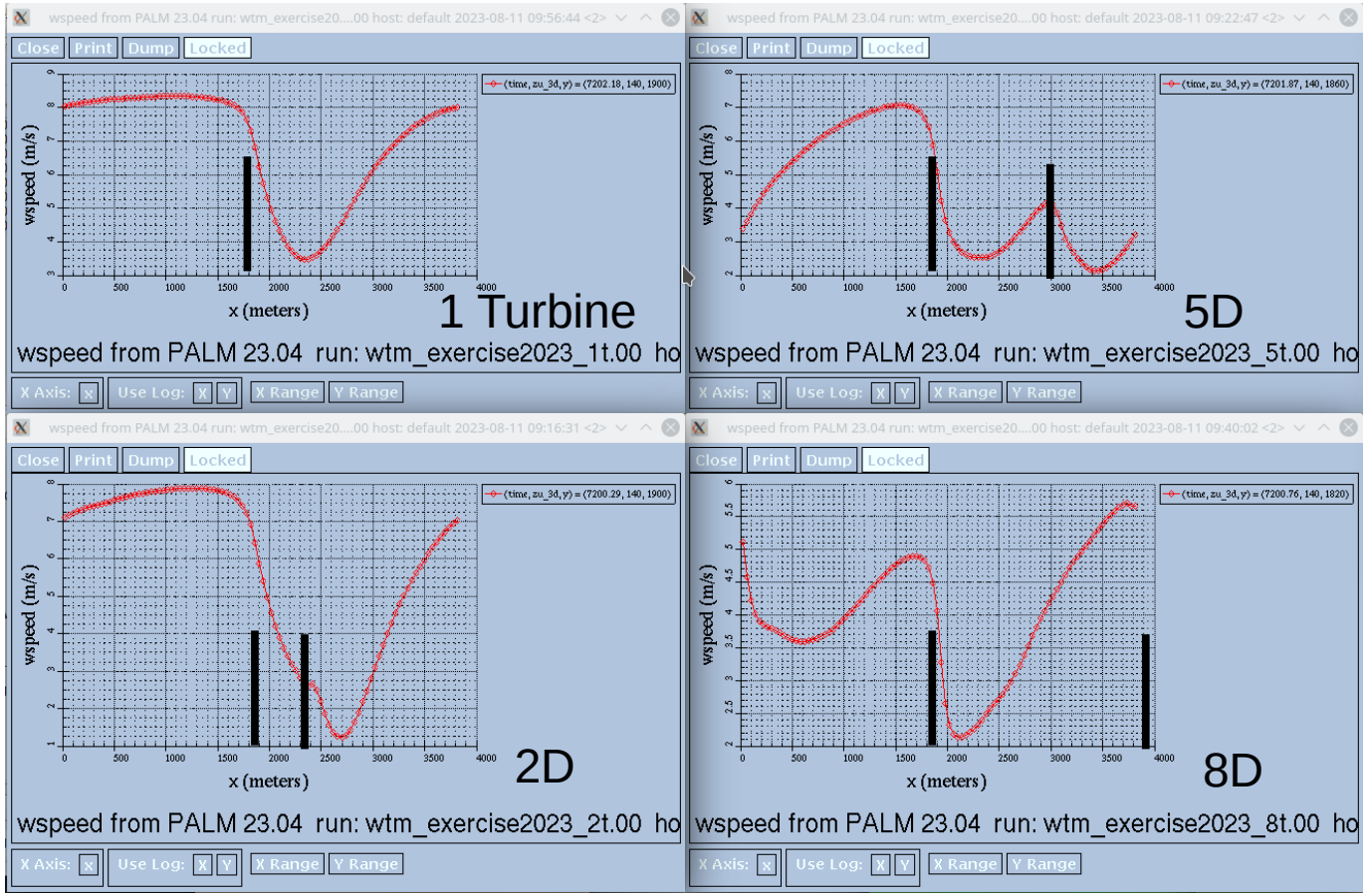
Answers

5. How do the turbines interact regarding wind speed recovery?

Flow recovery



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Answers

5. How do the turbines interact regarding wind speed recovery?

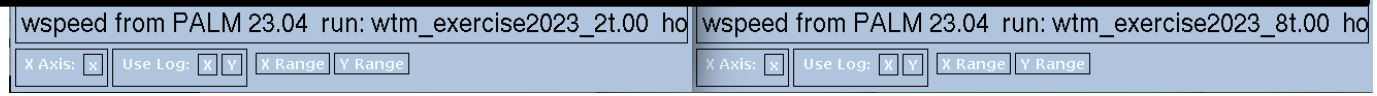
Flow reco

Results

2D
- the flow is not recovered when it approaches the 2nd turbine

5D
- the flow is partially recovered when it approaches the 2nd turbine
- The wake of the 2nd turbine affects the first turbine as well

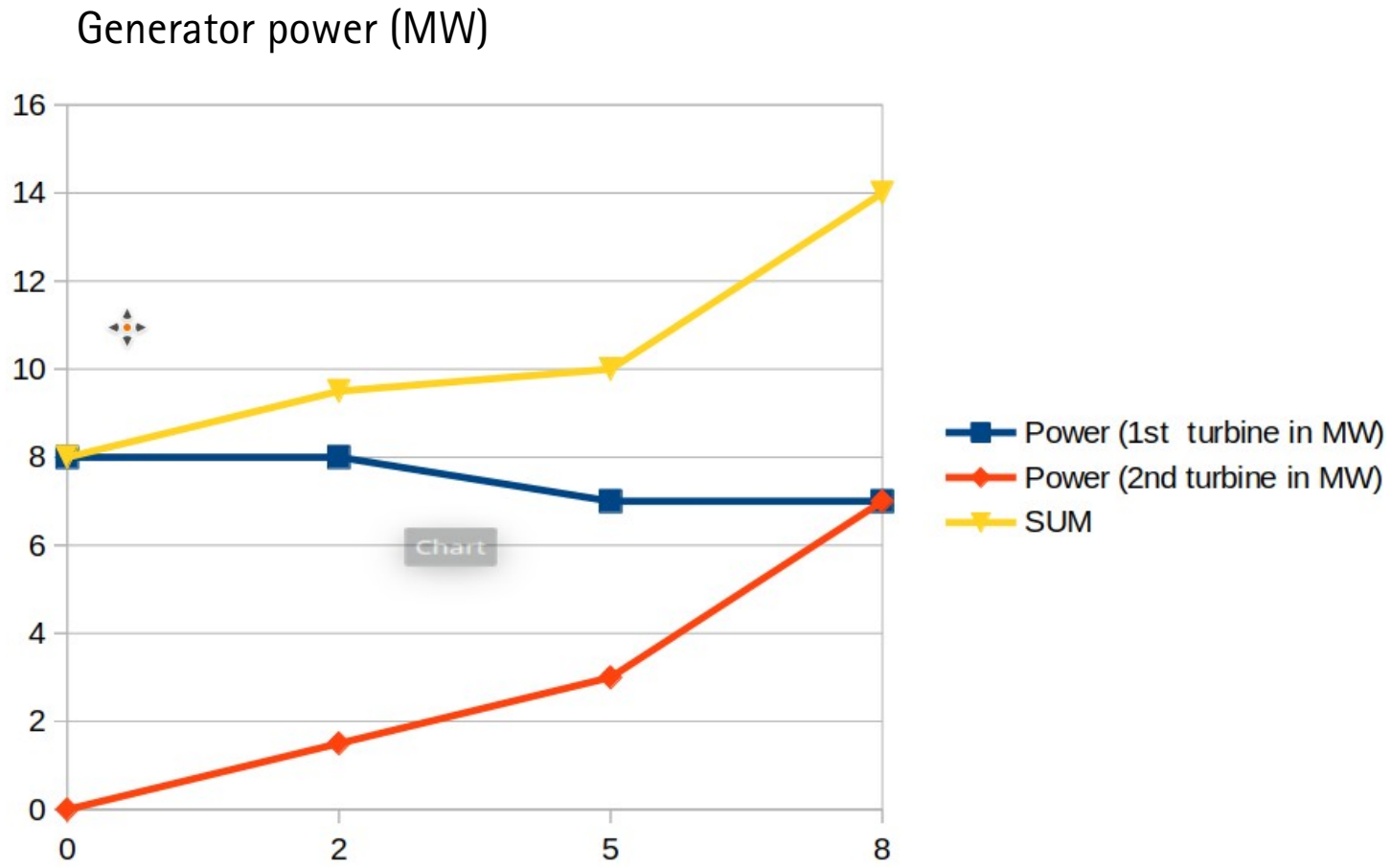
8D
- The flow is mostly recovered after the turbines (some differences persist, though due to unsteady flow)



E5: Wind turbine model

Answers

5. How do the turbines interact regarding wind speed recovery?



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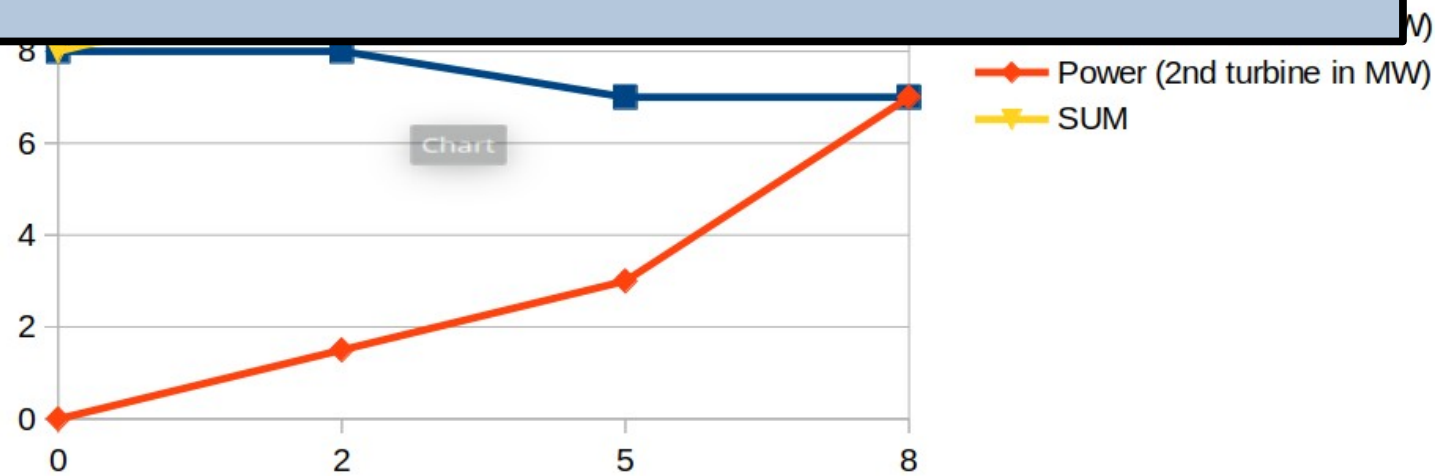
Answers

5. What distance between turbines do you recommend based on these lightweight simulations?

Results:

- Power yield of turbine 1 suffers from wake effects of turbine 2
- For 8D, power yield is about 14 MW (compared to 8 MW for a single turbine)

→ A distance of ~ 8D is recommended!